BachelorThesis

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1 Introduction

The problem of estimating species richness, which is the total number of distinct species present in an assemblage, including those not observed in any sample, has a long history across multiple disciplines. The 1943 paper by Fisher, Corbet and Williams [**?**] laid the mathematical foundation for statistical approaches to this problem. Around the same time, Good and Turing developed frequency estimation methods as part of their effort to crack German ciphers during World War II [**?**, **?**]. Since then the "species problem" has appeared in ecology (estimating butterfly species on an island), biology (counting microbial taxa), genetics (enumerating alleles), software engineering (finding undiscovered bugs), and linguistics (determining how many words an author knew) [**?**, **?**, **?**].

In nearly all of these settings, counting the true richness is almost impossible. A sample will almost always miss a fraction of the true species, and the simple count of observed species S_{obs} systematically underestimates the true richness S. The magnitude of this negative bias depends on the heterogeneity of the underlying abundance distribution: the more unequal the species abundances, the more rare species the sample will fail to detect. This makes the gap between S_{obs} and S the central statistical challenge of the field.

Our main question throughout this thesis is: *how can we estimate the total spoken vocabulary of a person from a finite recording of their speech?* In this setting, each distinct word corresponds to a species and each token (word occurrence) corresponds to an individual. Later on we will define more precisely what we mean by a distinct word. Spoken vocabulary falls right into the regime where species richness estimation is most difficult: word frequencies are highly heterogeneous, governed by Zipf's law, which states that the frequency of a word is approximately inversely proportional to its rank [**?**]. A small number of words (*the*, *is*, *a and*) represent a large share of all tokens, while the majority of distinct words appear only once or twice in any finite sample. Therefore, the number of words observed exactly once or twice, commonly referred to as singletons and doubletons, is always large, and the fraction of the vocabulary that remains unseen is substantial even in lengthy recordings.

A person's *active* spoken vocabulary (with how many different words they speak on a daily basis) is considerably smaller than their *passive* vocabulary (the words they can understand). Research in applied linguistics suggests that comprehension of spoken texts requires knowledge of roughly 6 000 to 7 000 word families, whereas written texts require 8 000 to 9 000 [illegible]. But what a person *produces* in conversation is a much smaller, context-dependent subset of even their active vocabulary. Estimating the size of this productive vocabulary from a finite speech sample is therefore a nontrivial statistical problem, and one that is well suited to the tools of nonparametric species richness estimation. We will also compare estimates derived from spoken data with those from written text, where we expect greater lexical diversity due to the wider range of registers and the time available for word choice.

In this thesis we proceed as follows. In Section 2 we establish the mathematical framework and notation for species richness estimation, distinguishing abundance data from incidence data and defining sample coverage. Section 3 reviews the classical nonparametric estimators such as Good-Turing, Chao1, iChao1, ACE, and the jackknife, discussing the assumptions, strengths, and limitations of each. Section 4 presents the main theoretical contribution: shape-constrained estimation via k-monotone and completely monotone distributions, We explain why using the completely monotone model is a natural next step when the degree of monotonicity k is large. Section 5 describes our data sources: the Shakespeare word counts from Spevack [illegible] replication, and our own spoken language corpus for the novel analysis. Section 6 presents the results, including estimator comparisons, rarefaction and extrapolation curves. We discuss the findings in Section 7 and conclude in Section 8.

2 Framework and Notation

Let us define our framework and notation. Consider an assemblage consisting of N total individuals, each belonging to one of S distinct species. Let N_i denote the number of individuals of the ith species in the assemblage, with $i = 1, 2, \ldots, S$ and $N_i > 0$. The total population size is

$$N = \sum_{i=1}^{S} N_i \,.$$

The relative abundance of species i is $p_i = N_i/N$, with $\sum_{i=1}^{S} p_i = 1$. The quantities N, S, N_i, and p_i represent the true and unknown parameters of the assemblage.

Assume we draw a random sample of n individuals from the assemblage. In this sample, a total of S_{obs} distinct species are observed, we call it the *reference sample*. Let X_i be the number of individuals of the ith species observed in this sample. Only species with $X_i > 0$ are observable, and $\sum_{i=1}^{S} X_i = n$. We denote

by f_k the *abundance frequency count*: the number of species represented by exactly k individuals in the reference sample, for $k = 1, 2, \ldots, n$. That is,

$$f_k = \#\{i : X_i = k\}\,.$$

The sample size can then be expressed in terms of frequency counts as

$$n = \sum_{i=1} X_i = \sum_{k \geq 1} k\, f_k\,.$$

We call f_1 the number of *singletons* (species observed exactly once) and f_2 the number of *doubletons* (species observed exactly twice). Singletons and doubletons play a central role in many estimation methods, as we will see in Section 3. More generally, most of the information about the undetected species is contained in the lower-order frequency counts $f_1, f_2, \ldots, f_r$ for small r. The quantity we wish to estimate is f_0: the number of species present in the assemblage but absent from the reference sample. The true species richness decomposes as

$$S = S_{\text{obs}} + f_0\,. \tag{1}$$

Since S_{obs} is directly observed, the entire estimation problem reduces to estimating f_0.

2.1 Sampling data structures

There are two principal types of sampling data: abundance data and incidence data.

Abundance data. In abundance sampling, the sampling unit is an individual. In the discrete-type variant, a fixed number n of individuals is drawn from a study area. In the continuous-type variant, sampling effort is measured on a continuous scale such as time, area, or volume. For our application, we record a speaker for a fixed period of time, and each word said constitutes an individual drawn from the speaker's vocabulary. The resulting data are discrete-type abundance data.

Incidence data. In incidence sampling, the sampling unit is not an individual but a spatial or temporal unit (a quadrat, a time window, etc.). For each of T sampling units, one records only the detection or non-detection of each species, yielding an incidence matrix $[W_{ij}]$ and incidence-based frequency counts Q_k ($k = 1, \ldots, T$), the number of species detected in exactly k sampling units. Since our data consist of individual words with their frequencies, we work exclusively with abundance data throughout this thesis. For a detailed treatment of incidence-based methods, we refer to Chao and Chiu [

2.2 Sample coverage

An important concept for assessing how complete a sample is, is the notion of *sample coverage*. Introduced by Good and Turing [**?**], it estimates the proportion of the total number of individuals in the assemblage that belong to species represented in the sample. Turing's estimator is

$$\hat{C} = 1 - \frac{f_1}{n} \,. \tag{2}$$

Intuitively, f_1/n estimates the probability that the next sampled individual belongs to a species not yet observed, subtracting it from 1 gives the coverage. A coverage close to 1 indicates that most of the assemblage's individuals belong to species already detected, whereas a low coverage signals that many species remain unobserved.

Chao and Jost [**?**] proposed a bias-corrected version:

$$\hat{C} = 1 - \frac{f_1}{n} \cdot \frac{(n-1)\, f_1}{(n-1)\, f_1 + 2\, f_2} \,. \tag{3}$$

This adjustment reduces the bias of the basic Turing estimator, particularly when singletons dominate the sample (i.e. when f_1 is large relative to n). Sample coverage will play an important role later, both as a diagnostic for the completeness of our speech samples and as the basis for coverage-based standardization when comparing samples of different sizes.

2.3 Parametric and nonparametric approaches

A parametric model assumes that the species abundance distribution follows a known family of distributions, fully characterized by a finite parameter vector θ. Common choices include the log-normal, the inverse-Gaussian, and the Zipf-Mandelbrot law. When such models can yield efficient estimators when the assumed form is correct, simulations have shown that they perform poorly when it is not [**?**]. Moreover, two parametric models may fit the observed data equally well yet produce widely different estimates of S, making the choice of distributional family a source of considerable uncertainty.

A nonparametric model makes no assumption about the functional form of the underlying abundance distribution. Instead, the number of undetected species is estimated directly from the lower-order frequency counts $f_1, f_2, \ldots, f_r$ for some small r. This approach is more robust across applications and permits meaningful comparisons between assemblages, since it does not require that different assemblages follow the same distributional family. All the estimators considered in this thesis are nonparametric.

2.4 Rarefaction, extrapolation, and standardization

When comparing species richness across multiple assemblages (for example, the vocabulary of two different speakers, or spoken versus written language from the same individual), a direct comparison of S_{obs} is misleading if the sample sizes differ, since larger samples will generally detect more species. Two standardization approaches are commonly used.

Sample-size-based rarefaction and extrapolation. Rarefaction asks: if we had drawn only $m < n$ individuals, how many species would we expect to observe? By subsampling from the reference sample, one constructs a *rarefaction curve* that shows expected richness as a function of sample size. Conversely, extrapolation projects the curve beyond the observed sample size, predicting how many additional species would be detected if sampling effort were increased [**?**, **?**].

Coverage-based standardization. An alternative, proposed by Chao and Jost [**?**], standardizes samples not by size but by coverage. The rationale is that two samples of equal size may represent very different levels of completeness: a sample of $n = 5\,000$ tokens from a speaker with homogeneous word usage may achieve near-complete coverage, while the same n from a speaker with a highly heterogeneous vocabulary may miss a substantial fraction of species. By comparing richness at equal coverage, one obtains a fairer comparison that accounts for the difficulty of each assemblage. We will use both approaches when comparing our spoken and written data in Section 6.

3 Existing Nonparametric Estimators

In this section we review the principal nonparametric estimators of species richness. All of them estimate the number of undetected species f_0 using only the lower-order frequency counts, without assuming any particular distributional form for the species abundances.

3.1 Good–Turing frequency estimation

The Good–Turing frequency estimator, developed by I. J. Good based on ideas from Alan Turing, originated in their wartime work on cracking German ciphers [**?**, **?**]. Good initially modelled the species frequencies using a multinomial distribution, but found this approach inaccurate for rare species. The key insight of Good–Turing estimation is to adjust the observed frequency of a species: for a species observed k times, its estimated probability is

$$p_k = \frac{(k+1) \cdot f_{k+1}}{n \cdot f_k}\,, \tag{4}$$

where n is the total sample size. Setting $k = 0$ in this formula gives the estimated total probability mass of all unseen species:

$$\hat{p}_0 = \frac{f_1}{n} \,.$$

This quantity is *not* the count of undetected species f_0; rather, it estimates the fraction of the assemblage's individuals that belong to species absent from the sample. It is the complement of the sample coverage $\hat{C} = 1 - f_1/n$ defined in Section 2.2.

Simple Good–Turing smoothing. A practical difficulty with Equation (4) is that for large k, the frequency counts f_k become noisy and sparse, making the ratio f_{k+1}/f_k unstable. The Simple Good–Turing (SGT) algorithm addresses this by replacing the raw counts with smoothed values $S(f_k)$ obtained from a log-linear regression of $\log f_k$ on $\log k$ [**?**]:

$$p_k = \frac{(k+1) \cdot S(f_{k+1})}{n \cdot S(f_k)} \,. \tag{5}$$

For small values of k, where the counts are reliable, one uses $S(f_k) = f_k$ directly, the smoothed values are used only in the sparse tail. We will not go into further detail on the regression procedure here, as the SGT algorithm is not the main focus of our analysis.

3.2 Chao1-type estimators

The Chao1 estimator, introduced by Chao [**?**], is the most widely used non-parametric lower bound for species richness. Its derivation applies the Cauchy-Schwarz inequality to the expected frequency counts, which guarantees that the result is a lower bound for S regardless of the true abundance distribution. (MAYBE DERIVE IT COMPLETELY) The bound depends only on S_{obs}, f_1, and f_2:

$$\hat{S}_{\text{Chao1}} = \begin{cases} S_{\text{obs}} + \dfrac{n-1}{n} \cdot \dfrac{f_1^2}{2f_2} \,, & \text{if } f_2 > 0 \,, \\ S_{\text{obs}} + \dfrac{n-1}{n} \cdot \dfrac{f_1(f_1-1)}{2} \,, & \text{if } f_2 = 0 \,. \end{cases} \tag{6}$$

For large sample sizes n, the finite-sample correction factor $(n-1)/n$ is negligible and the estimator simplifies to

$$\hat{S}_{\text{Chao1}} \approx \begin{cases} S_{\text{obs}} + \dfrac{f_1^2}{2f_2} \,, & \text{if } f_2 > 0 \,, \\ S_{\text{obs}} + \dfrac{f_1(f_1-1)}{2} \,, & \text{if } f_2 = 0 \,. \end{cases} \tag{7}$$

The Chao1 estimator achieves its lower bound with equality when all undetected species have the same detection probability as the observed singletons,

i.e. when the rare species are approximately homogeneous in their abundances. In practice, this condition is rarely met for linguistic data, where word frequencies are highly heterogeneous. When heterogeneity is high, the true richness substantially exceed the Chao1 estimate, meaning the estimator underestimates the true richness [**?**, **?**].

The iChao1 estimator. Chiu et al. [**?**] derived an improved lower bound by extending the Cauchy-Schwarz argument to incorporate tripletons f_3 and quadrupletons f_4:

$$\hat{S}_{i\text{Chao1}} = \hat{S}_{\text{Chao1}} + \frac{n-3}{n} \cdot \frac{f_3}{4f_4} \times \max\left(f_1 - \frac{n-3}{n-1} \cdot \frac{f_2 f_3}{2f_4},\ 0\right). \tag{8}$$

For large n, this simplifies to $\hat{S}_{i\text{Chao1}} \approx \hat{S}_{\text{Chao1}} + \frac{f_3}{4f_4} \max\left(f_1 - \frac{f_2 f_3}{2f_4},\ 0\right)$. By incorporating higher-order frequency counts, iChao1 tightens the lower bound when there is additional information in f_3 and f_4 beyond what f_1 and f_2 alone provide.

3.3 ACE-type estimators

The Abundance-based Coverage Estimator (ACE), developed by Chao and Lee [**?**], extends the idea of sample coverage by incorporating a measure of heterogeneity among the rare species. The key quantity is the squared coefficient of variation (CV) of the species abundances:

$$\gamma^2 = \frac{S^{-1} \cdot \sum_{i=1}^{S} (p_i - \bar{p})^2}{\bar{p}^2}\,, \tag{9}$$

where $\bar{p} = 1/S$. This characterizes the degree of heterogeneity among species abundances: $\gamma^2 = 0$ if and only if all species have equal abundances (homogeneous assemblage), and larger values of γ^2 indicate greater heterogeneity.

The central idea of ACE is that abundant species are almost certain to appear in any sample and therefore carry no information about the undetected fraction f_0. Only the rare species, whose detection is uncertain, are informative. ACE formalizes this by splitting the observed species into two groups using a cutoff value k (conventionally $k = 10$, following Chao and Lee [**?**]):

$$S_{\text{abun}} = \sum_{i>k} f_i\,, \qquad S_{\text{rare}} = \sum_{i=1}^{k} f_i\,,$$

$$n_{\text{rare}} = \sum_{i=1}^{k} i\, f_i\,, \qquad \hat{C}_{\text{rare}} = 1 - \frac{f_1}{n_{\text{rare}}}\,,$$

where n_{rare} is the sample size restricted to the rare group and $\hat{C}_{\text{rare}}$ is Turing's coverage applied to that group. The ACE estimator is then

$$\hat{S}_{\text{ACE}} = S_{\text{abun}} + \frac{S_{\text{rare}}}{\hat{C}_{\text{rare}}} + \frac{f_1}{\hat{C}_{\text{rare}}} \cdot \hat{\gamma}^2_{\text{rare}}\,, \tag{10}$$

with the estimated squared CV of the rare group given by

$$\hat{\gamma}^2_{\text{rare}} = \max\left\{ \frac{S_{\text{rare}}}{\hat{C}_{\text{rare}}} \cdot \frac{\sum_{i=1}^{k} i(i-1)\, f_i}{\left(\sum_{i=1}^{k} i\, f_i\right)\left(\sum_{i=1}^{k} f_i - 1\right)} - 1,\ 0 \right\}. \tag{11}$$

The cutoff $k = 10$ is conventional and simulation studies suggest that the estimator is reasonably robust to moderate changes in this value [**?**]. However, for data with extremely heavy tails (as in linguistic applications), the choice of cutoff may have a larger effect and must be studied.

3.4 Jackknife estimators

The jackknife is a general-purpose technique for reducing the bias of an estimator [**?**, **?**]. The idea is as follows: from the reference sample of n individuals, one systematically removes one individual at a time and recomputes the estimator on each reduced sample. The resulting *pseudo-values* are then combined to produce a bias-corrected estimate. This is the first-order jackknife; the second-order version extends the procedure by considering pairs of removed individuals, yielding a more aggressive bias correction at the cost of higher variance.

The first-order jackknife estimator for species richness is

$$\hat{S}_{jk1} = S_{\text{obs}} + \frac{n-1}{n} \cdot f_1 \approx S_{\text{obs}} + f_1\,, \tag{12}$$

and the second-order jackknife estimator is

$$\hat{S}_{jk2} = S_{\text{obs}} + \frac{2n-3}{n} \cdot f_1 - \frac{(n-2)^2}{n(n-1)} \cdot f_2 \approx S_{\text{obs}} + 2f_1 - f_2\,. \tag{13}$$

The first-order jackknife adds approximately f_1 to S_{obs}, effectively assuming that every singleton represents one missed species. This is a rough correction: it tends to underestimate the true richness when the sample is small (because many species are missed entirely and do not even appear as singletons) and to overestimate when the sample is large (because some singletons are genuinely rare species that would not yield additional undetected species). The second-order jackknife partially addresses this by subtracting a term proportional to f_2, which tempers the correction when doubletons are numerous [**?**, **?**].

3.5 Summary and limitations

Each of the estimators above has characteristic strengths and weaknesses. Good–Turing provides the conceptual foundation (sample coverage and the probability mass of unseen species) but does not directly yield an estimate of f_0 as a count. Chao1 gives a guaranteed lower bound that is easy to compute and widely applicable, but it can severely underestimate the true richness when species abundances are highly heterogeneous, precisely the situation we face with word frequency data. iChao1 tightens the bound by incorporating f_3 and f_4, but

remains a lower bound. ACE accounts for heterogeneity through the coefficient of variation, but depends on the choice of cutoff k. The jackknife estimators offer bias reduction but lack the theoretical guarantee of a lower bound and can overshoot.

A common limitation shared by all of these methods is that they use only the first few frequency counts and treat the rest of the abundance distribution as a black box. In the next section, we explore a fundamentally different approach: instead of extracting information from a handful of frequency counts, we impose a *shape constraint* on the entire distribution of abundances. This leads to the k-monotone and completely monotone estimators, which can exploit more of the available information while still remaining nonparametric.

4 Beyond Frequency-Count Estimators

The classical estimators discard most of the obeserved abundance distribution and in this section we will look into models that make full use of it.

Main part, look in depth on how the complete monotonicity works
Look how the results behave under different assumptions

- Parametric mixture models - how it works
- Frequency-ratio regression (Breakaway)
- Shape-constrained estimation:
- The mixture model perspective: why the distribution of abundances should be a mixture. Poisson mixtures and their boundary problem.
- k-monotone distributions: definition, the spline representation, the inversion formula
- The "one cannot see what cannot be seen" principle: how $\pi_0^k = 0$ identifies $p_0(0)$
- The Chee & Wang (2016) approach: k-monotone MLE as an alternative to Poisson mixtures, the problem of selecting k,
- Complete monotonicity as the limit of k-monotonicity (Steutel's theorem: completely monotone pmf $\leftrightarrow$ mixture of geometrics)
- The Balabdaoui & Kulagina estimator: the logarithmic choice of k_n, the estimator $\hat{N_\infty}$, asymptotic normality (state the result, explain the assumptions, confidence interval formula)
- The bootstrap test for complete monotonicity
- Brief numerical illustration on a known dataset

5 Data and Methodology

- What is a word, (ex: are run and running to different or not)
- what language are we working with: comparison of Australian english, American english and British english. But also cross language comparison.
- When applying it to German, the compound words will need to be separated.
- Deal with filler words such as "um", "hmmm" etc...
- how do we get the data: with existing recordings and/or new self-made recordings.
- transcript method and preprocessing.
- we will also apply our models to existing datasets to have a baseline: Shakespeare dataset for example.
- code the models we want to use (all of them maybe) - Implementation

6 Results

- replicate Shakespeare analysis to have a baseline
- apply other estimators
- estimator comparisons
- rarefaction/extrapolation curves

7 Discussion

8 Conclusion

Summarize findings, answer research questions, suggest future work.